File Edit Format Run Options Window Help

```
import heapq

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

heuristic = {
    'A': 7,
    'B': 6,
    'C': 4,
    'D': 5,
    'E': 3,
    'F': 2,
    'G': 0
}

def best_first_search(start, goal):
    queue = [(heuristic[start], start, [start])]
    visited = set()

    print("BEST FIRST SEARCH ALGORITHM")
    print("Starting City:", start)
    print("Destination City:", goal)
    print("-" * 50)

    while queue:
        h, node, path = heapq.heappop(queue)

        if node in visited:
            continue

        visited.add(node)

        print("Current City:", node)
```

File Edit Shell Debug Options Window Help

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AI.py ==========
========
BEST FIRST SEARCH ALGORITHM
Starting City: A
Destination City: G
--------------------------------------------------
Current City: A
Heuristic Value: 7
Path so far: ['A']
Visited Cities: ['A']
--------------------------------------------------
Adding City: B
Heuristic Value: 6
Adding City: C
Heuristic Value: 4

Current City: C
Heuristic Value: 4
Path so far: ['A', 'C']
Visited Cities: ['A', 'C']
--------------------------------------------------
Adding City: F
Heuristic Value: 2
Adding City: G
Heuristic Value: 0

Current City: G
Heuristic Value: 0
Path so far: ['A', 'C', 'G']
Visited Cities: ['A', 'G', 'C']
--------------------------------------------------
DESTINATION REACHED!
Final Path: A -> C -> G
Total Cities Visited: 3
Total Steps: 2
Heuristic Value at Destination: 0
>>>
```